

Ordered Hurwitz Resonances and Cubic Stieltjes Products

Exact Laurent coefficients, Gamma generating functions,
sharp curve dependence, and a polylogarithmic Mellin bridge

Research continuation prepared for Vladimir Reshetnikov
Mathematical derivations and independent review with OpenAI ChatGPT

10 October 2026

Abstract

We answer the ordered-depth-three and tangential-approach questions in the recent ProveIt resonant-jets programme, and obtain a stronger exact evaluation family at every depth. For a positive common shift, an elementary Hurwitz-tail subtraction constructs the canonical ordered Laurent germs with absolutely convergent harmonic-Stieltjes coefficients. Their diagonal regular generating function is explicit, and at zero deformation it is the Gamma quotient $\Gamma(a)/\Gamma(a+y)$. It follows that the constant Laurent coefficient of $Z_a(1+p\varepsilon, 1+q\varepsilon, \dots, 1+q\varepsilon; a)$ is a finite expression in ordinary Hurwitz derivatives and generalized Stieltjes constants, for every depth and every transverse pair of slopes. Explicit formulas are given through depth five. At depth three, arbitrary independent slopes require only one additional coefficient, identified precisely with a harmonic Stieltjes constant.

For analytic curves with prefix vanishing orders $\nu_1, \dots, \nu_d$, we prove an exact finite coefficient formula and show that the curve jet through order $\sum \nu_j + \max \nu_j$ is both sufficient and, in the worst case, necessary. A separate endpoint calculation constructs all cubic coincident Stieltjes moments, including arbitrary argument derivatives. The specified finite part of $\int_0^1 \psi(x)^3 dx$ is reduced to one diagonal Tornheim third derivative, with an independent polylogarithmic Mellin representation and numerical check.

The general ordered Laurent framework and the triple-Hurwitz/Tornheim correspondence have established antecedents and are credited explicitly. No arithmetic independence or first-publication claim is made. The short Gaussian S_6 and current S_8 identities remain conjectural, and reduction of the remaining Tornheim derivative to depth-one constants remains open here.

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1 Research targets, results, and source status

The question motivating this continuation is how much of a singular *ordered* nested sum can be evaluated using the familiar Gamma, polygamma, Hurwitz-zeta and Stieltjes functions. Full symmetrization removes ordering information and often reduces by the stuffle product. The individual ordered germ retains that information. Its singularities also meet along several hyperplanes, so a constant Laurent coefficient must specify the approach to their intersection.

We work from the repository snapshot

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The source survey includes the canonical manuscript, its status records, and the seven ZIP packages then present in `docs/incoming`. The mathematical audit concentrates on the relevant ordered-germ, coincident-product, resonance, and Gaussian-status material; it is not a re-verification of every proof in the full manuscript. The exact files and archive hashes are recorded in the accompanying `integration/source_snapshot.json`. All sources were inspected as research evidence; successful numerical replay is not treated as a proof of an analytic theorem.

1.1 The concrete questions answered

The incoming *Resonant Gamma Jets and Directional Zeta Identities* report asks for an ordered depth-three germ with independent exponents, its intersecting polar divisors, a normally convergent remainder, and the constant along every transverse ray. It separately asks for tangential curves and the exact number of curve coefficients needed to determine a finite part [2, Further research, Q1 and Q3]. Both questions are answered here. The first answer also yields the following all-depth specialization:

$$\text{FP}_{\varepsilon=0} Z_d(1 + p\varepsilon, 1 + q\varepsilon, \dots, 1 + q\varepsilon; a), \quad a > 0. \quad (1.1)$$

It closes in ordinary Hurwitz-zeta derivatives and Stieltjes constants whenever all prefix slopes $p + jq$ are nonzero. The initial slope may differ from every trailing slope; no symmetrization of the ordered sum is needed.

The *Coincident Stieltjes* report asks for a cubic closure theorem and specifically for the coordinate-normalized digamma cube [3, Further research: cubic coincident moments]. We construct the full cubic completion and evaluate that finite part in terms of a single precisely defined diagonal Tornheim derivative. This is a finite reduction to an established higher-depth function. A further reduction of that derivative to ordinary constants is a separate unanswered question.

Result	Exact outcome
Ordered local germs	Convergent remainder construction at every depth and every common positive shift; finite coefficient recursion.
One distinguished slope	Every nonpositive Laurent coefficient reduces to finitely many ordinary Hurwitz and Stieltjes jets.
Independent depth three	Complete pole part and constant term; one harmonic Stieltjes coefficient contains the residual ordering dependence.
Tangential curves	Pole order $\sum \nu_j$ and a sharp required curve jet order $\sum \nu_j + \max \nu_j$.
Cubic coincident products	All Stieltjes indices and all argument derivatives; explicit digamma-cube reduction to $\tau'''(0)$.

1.2 What is classical and what is added to this programme

Laurent expansions of Euler–Zagier multiple zeta functions have a substantial literature. Matsumoto, Onozuka and Wakabayashi analyze Laurent coefficients at integer points [6]. Saha constructs regularized multiple Stieltjes germs and, in particular, gives the canonical ordered decomposition at the all-one point [5, Theorem 2]. Our tail-defect proof realizes that structure for an arbitrary positive common Hurwitz shift. The explicit slope evaluations, the reduction of every transverse depth-three directional finite part to one named harmonic coefficient and ordinary Hurwitz data, and the sharp curve-jet statement are the specific conclusions developed for the repository’s current questions.

The harmonic Hurwitz-zeta convention is that of Kargın, Dil, Cenkci and Can [7]; the strict ordered convention used here differs from their inclusive sum by a single Hurwitz-zeta term. We give the conversion rather than silently identifying the constants. The triple-Hurwitz Fourier relation is part of the Tornheim literature of Espinosa and Moll [8]. The Mellin method follows the established approach discussed by Borwein and Dilcher and by Bailey and Borwein [9, 10]. Low diagonal derivatives are recovered and used as consistency checks, without claiming their first evaluation. The bibliography is a targeted source comparison, not an exhaustive proof of publication priority.

1.3 Conventions and mathematical status

For $a > 0$ we use real logarithms of $n + a$, and

$$\zeta(1 + u, a) = \frac{1}{u} + \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(a) u^m, \quad \gamma_0(a) = -\psi(a). \quad (1.2)$$

A prime on ζ denotes its first, spectral derivative. For $m \geq 2$,

$$\zeta(m, a) = \frac{(-1)^m}{(m-1)!} \psi^{(m-1)}(a). \quad (1.3)$$

These conventions follow the standard Hurwitz calculus [13]. Without a shift, $\gamma_m = \gamma_m(1)$ and $\gamma = \gamma_0(1)$. The rising factorial is $(z)_k = \prod_{j=0}^{k-1} (z + j)$, with $(z)_0 = 1$. Brackets $[u^j]$ or $[y^j t^k]$ mean ordinary Taylor coefficients, so conversion to derivatives requires the displayed factorials.

For a specified meromorphic function of one variable, FP means its coefficient of ε^0 . For a spatial endpoint integral, it means the constant remaining in the cutoff expansion in the stated spatial coordinate. These two operations are related by explicit corrections in Section 5; they are not interchangeable definitions. The holomorphic regular germ is a third object, defined by an exact subtraction recursion.

The canonical manuscript already proves S_4 and retains the short S_6 and revised S_8 candidates as conjectures [1]. No argument in this article changes that status. We do not infer period independence, minimal arithmetic bases, or nonreducibility from a failed search. Every theorem below is proved analytically; the attached computations check finite algebra and independently evaluated examples.

2 Ordered Hurwitz germs and a family closing in ordinary zeta jets

Fix $a > 0$. All logarithms of $n + a$ are real. Write

$$Z_d(s_1, \dots, s_d; a) = \sum_{n_1 > \dots > n_d \geq 0} \prod_{j=1}^d (n_j + a)^{-s_j}, \quad Z_0 = 1.$$

The defining series converges normally whenever $\Re(s_1 + \dots + s_j) > j$ for $1 \leq j \leq d$. Put $u_j = s_j - 1$ and $P_k = u_1 + \dots + u_k$. Empty products equal one.

2.1 An elementary holomorphic remainder construction

Define the tail defect

$$D(s, x) = \zeta(s, x + 1) - \frac{x^{1-s}}{s-1}, \quad x > 0. \quad (2.1)$$

Its apparent singularity at $s = 1$ is removable. On $\Re s > 0$ it has the absolutely convergent representation

$$D(s, x) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-xt} \left(\frac{1}{e^t - 1} - \frac{1}{t} \right) dt. \quad (2.2)$$

Initially this follows by subtracting the usual Mellin integrals in $\Re s > 1$. The bracket is bounded near zero and is $O(1/t)$ at infinity, proving continuation and normal convergence in $\Re s > 0$. In particular, uniformly for s on a compact subset there, $D(s, x) = O(x^{-\Re s})$ as $x \rightarrow \infty$.

For $d \geq 2$, define

$$E_d(u_1, \dots, u_d; a) = \sum_{n_2 > \dots > n_d \geq 0} D(1 + u_1, n_2 + a) \prod_{j=2}^d (n_j + a)^{-1-u_j}. \quad (2.3)$$

This is holomorphic on the ℓ^1 ball

$$\mathcal{U}_d = \{\mathbf{u} \in \mathbb{C}^d : |u_1| + \dots + |u_d| < 1/2\}.$$

Indeed, on a compact subset the inner finite sum over $n_3, \dots, n_d$ is bounded by a constant times $(1+n_2)^{\sum_{j=3}^d |u_j|} (1+\log(1+n_2))^{d-2}$. The remaining outer summand is therefore $O((1+n_2)^{-2+\sum |u_j|} (1+\log(1+n_2))^{d-2})$. This is summable uniformly on compact subsets. Holomorphy follows by normal convergence; fixed spectral derivatives can also be taken termwise.

Set $R_0 = 1$ and

$$R_1(u; a) = \zeta(1 + u, a) - \frac{1}{u}.$$

Recursively, for $d \geq 2$, put

$$R_d(u_1, \dots, u_d; a) = E_d(u_1, \dots, u_d; a) + \frac{R_{d-1}(u_1 + u_2, u_3, \dots, u_d; a) - R_{d-1}(u_2, u_3, \dots, u_d; a)}{u_1}. \quad (2.4)$$

The last quotient is its holomorphic divided difference at $u_1 = 0$; equivalently, it equals the integral from zero to one of the first partial derivative of R_{d-1} at $(u_2 + \tau u_1, u_3, \dots, u_d)$. Thus every R_d is holomorphic on $\mathcal{U}_d$.

Theorem 2.1 (Canonical shifted ordered decomposition). *As meromorphic germs on $\mathcal{U}_d$,*

$$\boxed{Z_d(1 + u_1, \dots, 1 + u_d; a) = \sum_{k=0}^d \frac{R_{d-k}(u_{k+1}, \dots, u_d; a)}{P_1 P_2 \dots P_k}.} \quad (2.5)$$

The local polar hyperplanes are precisely $P_k = 0$, each generically simple in the independent prefix coordinates.

Proof. Summing first over $n_1 > n_2$ in the absolute-convergence domain gives

$$Z_d(1 + \mathbf{u}; a) = \frac{1}{u_1} Z_{d-1}(1 + u_1 + u_2, 1 + u_3, \dots, 1 + u_d; a) + E_d(\mathbf{u}; a).$$

Apply the induction hypothesis to the depth $d - 1$ function. Every term except its holomorphic remainder already has the required denominator $P_1 \dots P_k$. Splitting the remaining numerator into $R_{d-1}(u_2, \dots, u_d)$ plus its difference gives exactly (2.4). Normal convergence and the identity theorem then establish the meromorphic identity throughout $\mathcal{U}_d$. In the coordinates $P_1, \dots, P_d$, every displayed denominator is square-free. The coefficient of $P_1^{-1} \dots P_d^{-1}$ is one, because only the $k = d$ term contributes. Consequently no one of these polar divisors is removable as a germ. $\square$

For $a = 1$, the shape of (2.5) is the ordered Laurent decomposition of Saha[5, Theorem 2]. The construction above provides a short proof for arbitrary common shift $a > 0$ and supplies absolutely convergent representatives for every regular coefficient. The canonical framework is credited to that work; the explicit shifted construction and the evaluations below are derived here.

2.2 Explicit coefficients in harmonic–Stieltjes coordinates

Let

$$Q_m(x) = \gamma_m(x + 1) + \frac{\log^{m+1} x}{m + 1}.$$

In particular $Q_0(x) = \log x - \psi(x + 1)$, and

$$D(1 + u, x) = \sum_{m \geq 0} \frac{(-1)^m}{m!} Q_m(x) u^m.$$

Consequently, for a multi-index $\mathbf{m} = (m_1, \dots, m_d)$,

$$[\mathbf{u}^{\mathbf{m}}] E_d(\mathbf{u}; a) = \frac{(-1)^{|\mathbf{m}|}}{m_1! \dots m_d!} \sum_{n_2 > \dots > n_d \geq 0} \frac{Q_{m_1}(n_2 + a) \log^{m_2}(n_2 + a)}{n_2 + a} \prod_{j=3}^d \frac{\log^{m_j}(n_j + a)}{n_j + a}. \quad (2.6)$$

These series are absolutely convergent. For a fixed outer index they are finite nested generalized harmonic sums with logarithmic insertions. If $r_{\mathbf{m}}^{(d)} = [\mathbf{u}^{\mathbf{m}}] R_d$, then

$$r_{m_1, \dots, m_d}^{(d)} = [\mathbf{u}^{\mathbf{m}}] E_d + \binom{m_1 + m_2 + 1}{m_1 + 1} r_{m_1 + m_2 + 1, m_3, \dots, m_d}^{(d-1)}. \quad (2.7)$$

This is a finite-depth, exact coefficient algorithm, with base $r_m^{(1)} = (-1)^m \gamma_m(a)/m!$.

2.3 The diagonal regular germ and its Gamma normalization

Write $r_d(t; a) = R_d(t, \dots, t; a)$, with $r_0 = 1$. Define the generating function, formally in y and analytically for sufficiently small y, t ,

$$F_a(y, t) = \exp \left\{ y \left(\zeta(1+t, a) - \frac{1}{t} \right) + \sum_{m \geq 2} \frac{(-1)^{m-1}}{m} \zeta(m+mt, a) y^m \right\}. \quad (2.8)$$

Theorem 2.2 (Diagonal regular coefficients). *For every $d \geq 0$,*

$$r_d(t; a) = [y^d] F_a(y, t), \quad \sum_{d \geq 0} r_d(0; a) y^d = \frac{\Gamma(a)}{\Gamma(a+y)}. \quad (2.9)$$

Proof. For $\Re t > 0$, the product over the common lattice gives the standard elementary-symmetric generating series

$$\sum_{d \geq 0} Z_d(1+t, \dots, 1+t; a) y^d = \prod_{n \geq 0} \left(1 + \frac{y}{(n+a)^{1+t}} \right) = \exp \sum_{m \geq 1} \frac{(-1)^{m-1}}{m} \zeta(m+mt, a) y^m.$$

On the diagonal, (2.5) becomes $Z_d = \sum_{k=0}^d r_{d-k}(t)/(k!t^k)$. Thus multiplication by $\exp(-y/t)$ gives (2.8), coefficient by coefficient. Holomorphic continuation reaches $t = 0$. Finally, the Taylor series of $\log \Gamma(a+y)$ is

$$\log \Gamma(a) + \psi(a)y + \sum_{m \geq 2} \frac{(-1)^m}{m} \zeta(m, a) y^m.$$

Since $\gamma_0(a) = -\psi(a)$, exponentiation proves the last assertion. $\square$

Let $B_d(a) = [y^d] \Gamma(a)/\Gamma(a+y)$. These are explicit Bell polynomials in digamma and polygamma values. For example, abbreviating $g = \gamma_0(a)$ and $z_m = \zeta(m, a)$,

$$\begin{aligned} B_1 &= g, & B_2 &= \frac{g^2 - z_2}{2}, \\ B_3 &= \frac{g^3 - 3gz_2 + 2z_3}{6}, \\ B_4 &= \frac{g^4 - 6g^2z_2 + 3z_2^2 + 8gz_3 - 6z_4}{24}, \\ B_5 &= \frac{g^5 - 10g^3z_2 + 15gz_2^2 + 20g^2z_3 - 20z_2z_3 - 30gz_4 + 24z_5}{120}. \end{aligned}$$

2.4 All depths with one distinguished slope

The finite part below means the coefficient of ε^0 after the stated one-variable substitution. It is not a value assigned at the intersection of polar divisors.

Theorem 2.3 (A continuum of exact ordered finite parts at every depth). *Let $d \geq 1$, and let $p, q \in \mathbb{C}$ satisfy $p + jq \neq 0$ for $0 \leq j \leq d-1$. Put $D_k(p, q) = \prod_{j=0}^{k-1} (p + jq)$. Then*

$$\begin{aligned} & \text{FP}_{\varepsilon=0} Z_d(1+p\varepsilon, 1+q\varepsilon, \dots, 1+q\varepsilon; a) \\ &= \boxed{B_d(a) + \sum_{k=1}^{d-1} \frac{q^k}{D_k(p, q)} [y^{d-k} t^k] F_a(y, t)}. \end{aligned} \quad (2.10)$$

Every expression on the right is a finite polynomial in ordinary Hurwitz zeta derivatives and generalized Stieltjes constants, with coefficients rational in p, q . It uses only $\gamma_j(a)$ with $0 \leq j \leq d-1$ and $\zeta^{(j)}(m, a)$ with $m \geq 2$, $j \geq 0$, and $m+j \leq d$. The case $q = 0$ is included and gives simply $B_d(a)$.

Proof. For $k \geq 1$, the tail in (2.5) is diagonal, and its denominator becomes $D_k(p, q)\varepsilon^k$. Its constant coefficient is therefore $q^k[t^k]r_{d-k}(t; a)/D_k(p, q)$. The $k = 0$ term contributes $R_d(0) = B_d$, and the $k = d$ term is a pure pole and contributes zero. The coefficient formula (2.8) proves the closure and the stated finite range of orders. $\square$

The same argument evaluates the complete polar part. More precisely, for $-d \leq \ell \leq 0$,

$$[\varepsilon^\ell]Z_d(1 + p\varepsilon, 1 + q\varepsilon, \dots; a) = \sum_{k=\max(0, -\ell)}^d \frac{q^{k+\ell}}{D_k(p, q)} [y^{d-k}t^{k+\ell}]F_a(y, t), \quad (2.11)$$

where $D_0 = 1$ and $0^0 = 1$. For $k = 0$ this formula is used only when $\ell = 0$, where its contribution is B_d . Denote the finite part in (2.10) by $\mathcal{F}_d(p, q; a)$. For $q \neq 0$ it depends only on $\lambda = p/q$:

$$\mathcal{F}_d(p, q; a) = B_d(a) + \sum_{k=1}^{d-1} \frac{[y^{d-k}t^k]F_a(y, t)}{(\lambda)_k}. \quad (2.12)$$

Thus its dependence on the distinguished slope ratio is an explicit finite inverse-factorial expansion. Formula (2.12) can have a meromorphic continuation in λ beyond the transverse directions; such a continuation of a coefficient must not be confused with restriction of the original multiple-zeta germ to a polar divisor.

This is an exact evaluation theorem, not a claim of algebraic independence or minimality of the resulting constants. The general ordered problem is more complicated: independent trailing slopes can retain nonsymmetric multiple Stieltjes coefficients.

2.5 Explicit depths two through five

Set $g_j = \gamma_j(a)$, $g = g_0$, and $z_m^{[j]} = \partial_s^j \zeta(s, a)|_{s=m}$; thus $z_m = z_m^{[0]}$. Then

$$\mathcal{F}_2 = B_2 - \frac{q}{p}g_1, \quad (2.13)$$

$$\mathcal{F}_3 = B_3 - \frac{q}{p}(gg_1 + z_2^{[1]}) + \frac{q^2g_2}{2p(p+q)}. \quad (2.14)$$

At depth four one obtains

$$\begin{aligned} \mathcal{F}_4 = B_4 + \frac{q}{p} \left(-\frac{g^2g_1}{2} + \frac{g_1z_2}{2} - gz_2^{[1]} + z_3^{[1]} \right) \\ + \frac{q^2}{p(p+q)} \left(\frac{g_1^2 + gg_2}{2} - z_2^{[2]} \right) - \frac{q^3g_3}{6p(p+q)(p+2q)}. \end{aligned} \quad (2.15)$$

At depth five,

$$\mathcal{F}_5 = B_5 + \frac{q}{p}A_{4,1} + \frac{q^2}{p(p+q)}A_{3,2} + \frac{q^3}{p(p+q)(p+2q)}A_{2,3} + \frac{q^4g_4}{24p(p+q)(p+2q)(p+3q)}, \quad (2.16)$$

where

$$\begin{aligned} A_{4,1} &= -\frac{g^3 g_1}{6} + \frac{g g_1 z_2}{2} - \frac{g^2 z_2^{[1]}}{2} + \frac{z_2 z_2^{[1]}}{2} - \frac{g_1 z_3}{3} + g z_3^{[1]} - z_4^{[1]}, \\ A_{3,2} &= \frac{g g_1^2}{2} + \frac{g^2 g_2}{4} - \frac{g_2 z_2}{4} + g_1 z_2^{[1]} - g z_2^{[2]} + \frac{3}{2} z_3^{[2]}, \\ A_{2,3} &= -\frac{g g_3}{6} - \frac{g_1 g_2}{2} - \frac{2}{3} z_2^{[3]}. \end{aligned}$$

These formulas can be checked exactly using Newton's identities, with independent formal symbols for every Hurwitz jet.

2.6 Arbitrary depth-three directions and the one extra constant

Define

$$\eta(a) = \frac{\gamma_2(a)}{2} - \sum_{n \geq 0} \frac{\gamma_1(n+a+1) + \frac{1}{2} \log^2(n+a)}{n+a}. \quad (2.17)$$

The summand is $O((1 + \log n)/n^2)$. The identities above imply

$$\eta(a) = [u]R_2(u, 0; a), \quad [v]R_2(0, v; a) = -\gamma_0(a)\gamma_1(a) - \zeta'(2, a) - \eta(a).$$

For the second identity use the depth-two stuffle law, which gives

$$R_2(u, v; a) + R_2(v, u; a) = R_1(u; a)R_1(v; a) - \zeta(2 + u + v, a).$$

Theorem 2.4 (Every transverse straight direction at depth three). *Let $c_1, c_2, c_3 \in \mathbb{C}$ with $c_1(c_1 + c_2)(c_1 + c_2 + c_3) \neq 0$. Then*

$$\begin{aligned} &\text{FP}_{\varepsilon=0} Z_3(1 + c_1\varepsilon, 1 + c_2\varepsilon, 1 + c_3\varepsilon; a) \\ &= B_3(a) - \frac{c_3}{c_1}(\gamma_0(a)\gamma_1(a) + \zeta'(2, a)) + \frac{c_3^2 \gamma_2(a)}{2c_1(c_1 + c_2)} + \frac{c_2 - c_3}{c_1} \eta(a). \end{aligned} \quad (2.18)$$

The complete polar part is

$$\begin{aligned} &\frac{1}{c_1(c_1 + c_2)(c_1 + c_2 + c_3)\varepsilon^3} + \frac{\gamma_0(a)}{c_1(c_1 + c_2)\varepsilon^2} \\ &+ \frac{1}{\varepsilon} \left\{ \frac{B_2(a)}{c_1} - \frac{c_3 \gamma_1(a)}{c_1(c_1 + c_2)} \right\}. \end{aligned} \quad (2.19)$$

Proof. At depth three the canonical identity reads

$$Z_3 = \frac{1}{u_1(u_1 + u_2)(u_1 + u_2 + u_3)} + \frac{R_1(u_3)}{u_1(u_1 + u_2)} + \frac{R_2(u_2, u_3)}{u_1} + R_3(u_1, u_2, u_3).$$

Expand R_1 through quadratic order and R_2 through linear order; use $R_2(0, 0) = B_2$, $R_3(0, 0, 0) = B_3$, and the two displayed first coefficients of R_2 . This gives both assertions. $\square$

The coefficient of η vanishes exactly on the family $c_2 = c_3$ (without any assertion that η is independent of the other constants, or that it is nonzero at every shift). The full labelled symmetrization also cancels it, because the two orders of the last two slopes have opposite coefficients. Thus generic depth-three directional finite parts need only one nonsymmetric depth-two coefficient beyond ordinary Hurwitz data; no new depth-three constant is required for the constant Laurent term.

3 Tangential curves and the exact amount of curve data

Straight rays are only the simplest restrictions of the ordered germ. A curve can be tangent to one or several prefix hyperplanes, producing a higher pole order. The canonical decomposition gives a finite calculus for all such cases, including a sharp statement about which Taylor coefficients of the curve can affect the answer.

3.1 A coefficient formula for every admissible analytic curve

Let $u_i(\varepsilon)$ be analytic near zero, with $u_i(0) = 0$, and let

$$P_j(\varepsilon) = \sum_{i=1}^j u_i(\varepsilon). \quad (3.1)$$

Assume that no P_j is identically zero. There are unique integers $\nu_j \geq 1$ and analytic units p_j such that

$$P_j(\varepsilon) = \varepsilon^{\nu_j} p_j(\varepsilon), \quad p_j(0) \neq 0. \quad (3.2)$$

Set $M_0 = 0$ and $M_k = \nu_1 + \cdots + \nu_k$.

Theorem 3.1 (Exact curve finite part and sharp jet order). *The composed ordered function has a pole of exactly order M_d , with leading coefficient $\prod_{j=1}^d p_j(0)^{-1}$. Its constant term is*

$$\boxed{\text{FP}_{\varepsilon=0} Z_d(1 + u_1(\varepsilon), \dots, 1 + u_d(\varepsilon); a) = \sum_{k=0}^d [\varepsilon^{M_k}] \frac{R_{d-k}(u_{k+1}(\varepsilon), \dots, u_d(\varepsilon); a)}{p_1(\varepsilon) \cdots p_k(\varepsilon)}}. \quad (3.3)$$

Regular-germ Taylor coefficients through total degree M_d and curve Taylor coefficients through order

$$J = M_d + \max_{1 \leq j \leq d} \nu_j \quad (3.4)$$

suffice. For the class of curves with the specified prefix orders, the bound J cannot in general be decreased.

Proof. Insert (3.2) into (2.5). Its k th summand is ε^{-M_k} times a holomorphic function. The constant is therefore precisely the coefficient on the right of (3.3). Because $M_0 < M_1 < \cdots < M_d$, only the $k = d$ term can have pole order M_d . Its numerator is $R_0 = 1$, so its leading coefficient is nonzero and equals the asserted product.

In the k th numerator, each u_i vanishes at least to first order. Coefficients through order M_k thus use only regular-germ monomials of total degree at most M_k , and only curve coefficients through that order. To invert $p_j = P_j/\varepsilon^{\nu_j}$ through degree M_k requires P_j through degree $\nu_j + M_k$. These degrees are all at most (3.4), proving sufficiency.

For necessity, fix an index j and perturb only that prefix by

$$P_j(\varepsilon) \mapsto P_j(\varepsilon) + \delta \varepsilon^{\nu_j + M_d}. \quad (3.5)$$

This is realized by opposite changes to u_j, u_{j+1} when $j < d$, or by a change to u_d when $j = d$. All prefix orders and all other prefixes stay fixed. The affected unit first changes in degree M_d . Every

denominator change in a term with $k < d$ starts after its constant degree M_k ; numerator changes start even later. In the $k = d$ term, ordinary inversion gives the exact change

$$\Delta \text{FP} = -\frac{\delta}{p_j(0) \prod_{\ell=1}^d p_\ell(0)}. \quad (3.6)$$

Choosing j with maximal ν_j changes no curve coefficient below order J but changes the finite part by a nonzero multiple of δ . This proves sharpness. $\square$

The theorem does not assign a restriction to a curve lying identically in a polar hyperplane. Such a restriction needs an additional operation, for example first removing a specified polar germ. A finite algebraic value obtained by continuing a slope formula is not by itself that extra prescription.

3.2 Transverse curves and a depth-three fourth-jet formula

When every prefix is transverse, $\nu_j = 1$, Theorem 3.1 requires the curve jet through degree $d + 1$. Thus a quadratic curve description is insufficient even at depth two, and a cubic description is insufficient in general at depth three.

An explicit example is a common reparametrization of a fixed ray. Suppose

$$Z_3(1 + c_1 t, 1 + c_2 t, 1 + c_3 t; a) = A_{-3} t^{-3} + A_{-2} t^{-2} + A_{-1} t^{-1} + A_0 + O(t),$$

where all coefficients are given by (2.18)–(2.19). Put

$$t = h(\varepsilon) = \varepsilon + \alpha \varepsilon^2 + \beta \varepsilon^3 + \chi \varepsilon^4 + O(\varepsilon^5).$$

Then exact series inversion yields

$$\begin{aligned} \text{FP}_{\varepsilon=0} Z_3(1 + c_1 h(\varepsilon), 1 + c_2 h(\varepsilon), 1 + c_3 h(\varepsilon); a) \\ = A_0 - \alpha A_{-1} + (3\alpha^2 - 2\beta) A_{-2} \\ + (-10\alpha^3 + 12\alpha\beta - 3\chi) A_{-3}. \end{aligned} \quad (3.7)$$

For a transverse ray, $A_{-3} \neq 0$, so the fourth coefficient χ has an unavoidable effect. The identity follows by taking the constant coefficients of h^{-1}, h^{-2}, h^{-3} ; their respective values are $-\alpha$, $3\alpha^2 - 2\beta$, and $-10\alpha^3 + 12\alpha\beta - 3\chi$.

At the equal-slope ray $c_1 = c_2 = c_3 = 1$, one may insert

$$A_{-3} = \frac{1}{6}, \quad A_{-2} = \frac{\gamma_0(a)}{2}, \quad A_{-1} = \frac{\gamma_0(a)^2 - \gamma_1(a) - \zeta(2, a)}{2},$$

and $A_0 = \mathcal{F}_3(1, 1; a)$. This supplies a completely explicit curved depth-three identity using ordinary Hurwitz data.

3.3 A tangency example of arbitrarily high order

At depth two take

$$u_1(\varepsilon) = \varepsilon, \quad u_2(\varepsilon) = -\varepsilon + \varepsilon^r, \quad r \geq 2.$$

Here $(\nu_1, \nu_2) = (1, r)$, so the pole order is $r + 1$ and the sharp jet order is $2r + 1$. The canonical formula simplifies exactly to

$$Z_2 = \varepsilon^{-r-1} + \frac{R_1(-\varepsilon + \varepsilon^r; a)}{\varepsilon} + R_2(\varepsilon, -\varepsilon + \varepsilon^r; a).$$

Consequently

$$\text{FP } Z_2 = B_2(a) + \gamma_1(a). \quad (3.8)$$

Now change u_2 by $\delta\varepsilon^{2r+1}$. This leaves all lower curve coefficients unchanged, but the first displayed term contributes an additional constant $-\delta$. Thus the very short unperturbed answer does not reduce the amount of curve information needed in general.

3.4 Dependence on the ratio of straight slopes

For $q \neq 0$, put $\lambda = p/q$ and write $\mathcal{F}_d(\lambda; a) = \mathcal{F}_d(p, q; a)$. Theorem 2.3 gives the rational function

$$\mathcal{F}_d(\lambda; a) = B_d(a) + \sum_{k=1}^{d-1} \frac{[y^{d-k}t^k]F_a(y, t)}{(\lambda)_k}. \quad (3.9)$$

It has at most simple poles at $0, -1, \dots, 2-d$. In particular, for $d \geq 2$,

$$\boxed{\text{Res}_{\lambda=2-d} \mathcal{F}_d(\lambda; a) = -\frac{\gamma_{d-1}(a)}{(d-1)!(d-2)!}.} \quad (3.10)$$

Only the $k = d-1$ summand can contribute to this residue; its numerator is $(-1)^{d-1}\gamma_{d-1}(a)/(d-1)!$, and the remaining factors in the denominator have product $(-1)^{d-2}(d-2)!$. Equation (3.10) is an identity of the rational slope continuation. It is not a value of the original multiple zeta function on the excluded polar ray. It gives a useful inverse relation: a higher Stieltjes constant can be recovered from the slope dependence of the ordered finite-part family.

In fact the entire coefficient family has an exact inverse in terms of these residues. For $d \geq 2$, put

$$A_k = [y^{d-k}t^k]F_a(y, t), \quad \rho_j = \text{Res}_{\lambda=-j} \mathcal{F}_d(\lambda; a), \quad 0 \leq j \leq d-2.$$

Partial fractions in (3.9) give

$$\rho_j = \frac{(-1)^j}{j!} \sum_{k=j+1}^{d-1} \frac{A_k}{(k-1-j)!}, \quad \boxed{A_k = (-1)^{k-1} \sum_{j=k-1}^{d-2} \frac{j!}{(j-k+1)!} \rho_j.} \quad (3.11)$$

To verify the inverse, set $b_j = (-1)^j j! \rho_j$. The first transform is the upper-triangular convolution with $1/r!$; convolution with $(-1)^r/r!$ inverts it, because $e^z e^{-z} = 1$. This proves the second formula and recovers (3.10) at $k = d-1$.

3.5 Recovering every regular jet from directional finite parts

The one-distinguished-slope family determines diagonal combinations of regular coefficients. Allowing all independent slopes recovers each coefficient that can appear, with no loss of information.

Write $H_{r,m}(\mathbf{v}; a) = [t^m]R_r(t\mathbf{v}; a)$ for the homogeneous degree- m Taylor polynomial, and let

$$\mathcal{G}_d(\mathbf{c}; a) = \text{FP}_{\varepsilon=0} Z_d(1 + c_1\varepsilon, \dots, 1 + c_d\varepsilon; a).$$

For $P_j = c_1 + \dots + c_j \neq 0$, the canonical formula gives

$$\mathcal{G}_d(\mathbf{c}; a) = B_d(a) + \sum_{k=1}^{d-1} \frac{H_{d-k,k}(c_{k+1}, \dots, c_d; a)}{P_1 \cdots P_k}. \quad (3.12)$$

Corollary 3.2 (Exact reconstruction from directional data). *The rational function $\mathcal{G}_d$ uniquely determines every $H_{d-k,k}$, $1 \leq k \leq d-1$, and its constant B_d . In particular, the full collection of directional finite parts at all depths determines every regular germ R_r : its homogeneous degree- m part is recovered at depth $r+m$.*

Proof. Use the independent prefix coordinates $P_1, \dots, P_d$ and descend from $k = d-1$ to $k = 1$. Suppose all higher-index summands in (3.12) have been recovered and subtracted. Take the coefficient of $P_1^{-1} \cdots P_k^{-1}$ in the remainder, treating $P_{k+1}, \dots, P_d$ as generic parameters. The lower-index summands are holomorphic in at least one of these first k variables and contribute zero. The surviving coefficient is

$$H_{d-k,k}(P_{k+1}, P_{k+2} - P_{k+1}, \dots, P_d - P_{d-1}; a).$$

This invertible change of the remaining variables recovers the entire polynomial $H_{d-k,k}$. Subtracting all recovered terms leaves B_d . For a general R_r and $m \geq 1$, choose $d = r+m$ and $k = m$. The degree-zero part is $R_r(0) = B_r$, already known from the Gamma quotient. No higher Taylor degree than those in (3.12) can affect a finite part at the fixed depth. $\square$

4 The harmonic coefficient and its polylogarithmic representations

The coefficient $\eta(a)$ in (2.17) is not an undefined remainder. It has an absolutely convergent Stieltjes series, belongs to an established harmonic-zeta convention, and admits a convergent Mellin representation with an elementary kernel when $a = 1$. This section also records exact difference and derivative identities.

4.1 Strict and inclusive harmonic conventions

For $n \geq 0$ define the strict shifted harmonic sum

$$H_n^{(s)}(a) = \sum_{k=0}^{n-1} (k+a)^{-s}, \quad H_0^{(s)}(a) = 0.$$

It is entire in s for fixed n . Its spectral derivatives are finite logarithm-weighted harmonic sums. Summing first over the inner index gives

$$Z_2(s, 1; a) = \sum_{n \geq 0} \frac{H_n^{(1)}(a)}{(n+a)^s}. \quad (4.1)$$

The canonical decomposition with the second deviation zero yields

$$Z_2(1+u, 1; a) = \frac{1}{u^2} + \frac{\gamma_0(a)}{u} + B_2(a) + \eta(a)u + O(u^2). \quad (4.2)$$

Indeed the regular part is $R_2(u, 0; a)$ and its first coefficient is exactly (2.17).

Kargin et al. use the inclusive harmonic Hurwitz function [7]

$$\zeta_H(s, a) = \sum_{n \geq 0} (n + a)^{-s} \sum_{k=0}^n (k + a)^{-1}.$$

The relation and the coefficient conversion are

$$\zeta_H(s, a) = Z_2(s, 1; a) + \zeta(s + 1, a), \quad (4.3)$$

$$\begin{aligned} \gamma_H(0, a) &= \frac{\gamma_0(a)^2 + \zeta(2, a)}{2}, \\ \eta(a) &= \boxed{-\gamma_H(1, a) - \zeta'(2, a)}. \end{aligned} \quad (4.4)$$

Here $\gamma_H(m, a)$ is normalized by the coefficient $(-1)^m \gamma_H(m, a)/m!$ in the regular Laurent series of $\zeta_H(1 + u, a)$. This fixes both the strict/inclusive difference and the sign of the first coefficient.

4.2 Shift and parameter-derivative identities

Proposition 4.1 (Transport of the ordering coefficient). *For every $a > 0$,*

$$\boxed{\eta(a + 1) - \eta(a) = \frac{\gamma_1(a + 1)}{a}}. \quad (4.5)$$

Writing ∂_1, ∂_2 for the spectral derivatives of Z_2 , one also has the convergent identity

$$\begin{aligned} \eta'(a) &= -Z_2(2, 1; a) - \partial_1 Z_2(2, 1; a) + \partial_2 Z_2(2, 1; a) \\ &\quad + \gamma_1(a)\zeta(2, a) + \zeta'(3, a). \end{aligned} \quad (4.6)$$

All three double sums on the right converge absolutely, including their indicated logarithmic insertions.

Proof. Removing the terms with smallest index zero gives

$$Z_2(s, 1; a) - Z_2(s, 1; a + 1) = a^{-1}\zeta(s, a + 1).$$

The coefficient of $s - 1$ is $\eta(a) - \eta(a + 1) = -\gamma_1(a + 1)/a$, proving (4.5).

In an absolute-convergence domain differentiation in the common shift gives

$$\partial_a Z_2(s, 1; a) = -s Z_2(s + 1, 1; a) - Z_2(s, 2; a).$$

Use the stuffle identity

$$Z_2(s, 2; a) = \zeta(s, a)\zeta(2, a) - Z_2(2, s; a) - \zeta(s + 2, a)$$

and continue near $s = 1$. Taking the coefficient of $s - 1$ in (4.2) gives (4.6). Convergence follows from $H_n^{(1)}(a) = O(1 + \log n)$ and the extra outer square; spectral derivatives add only logarithmic factors. $\square$

For integer shifts, repeated use of (4.5) yields the finite identity

$$\eta(a + N) = \eta(a) + \sum_{k=0}^{N-1} \frac{\gamma_1(a + k + 1)}{a + k}.$$

These transport identities are consistent with the harmonic Hurwitz shift recurrence; they are included to make the ordering coefficient usable without a change of convention.

4.3 A convergent Lerch–polylogarithm Mellin formula

For $0 < z < 1$, let $\Phi(z, s, a) = \sum_{n \geq 0} z^n / (n + a)^s$. The harmonic generating function is

$$\sum_{n \geq 0} H_n^{(1)}(a) z^n = \frac{z\Phi(z, 1, a)}{1 - z}. \quad (4.7)$$

It follows by interchanging two absolutely convergent sums. Hence

$$\Gamma(s)Z_2(s, 1; a) = \int_0^\infty t^{s-1} K_a(t) dt, \quad K_a(t) = \frac{e^{-(a+1)t} \Phi(e^{-t}, 1, a)}{1 - e^{-t}}, \quad \Re s > 1. \quad (4.8)$$

Put $h_a = \gamma_0(a) - \gamma$ and define

$$\begin{aligned} J_a(s) &= \int_0^1 t^{s-1} \left(K_a(t) - \frac{-\log t + h_a}{t} \right) dt \\ &\quad + \int_1^\infty t^{s-1} K_a(t) dt. \end{aligned} \quad (4.9)$$

The kernel subtraction is $O(1 + |\log t|)$ at zero. Indeed, put $d_n = (n + a)^{-1} - (n + 1)^{-1}$. Then $d_n = O((n + 1)^{-2})$, $\sum d_n = h_a$, and

$$\sum_{n \geq 0} \frac{|1 - e^{-nt}|}{(n + 1)^2} = O(t(1 + |\log t|)).$$

Comparing $\Phi(e^{-t}, 1, a)$ with $\Phi(e^{-t}, 1, 1)$ proves the asserted estimate, locally uniformly for $a > 0$. Thus J_a is holomorphic for $\Re s > 0$, with all fixed spectral derivatives obtained by inserting powers of $\log t$. We have the exact continuation

$$\Gamma(s)Z_2(s, 1; a) = \frac{1}{(s - 1)^2} + \frac{h_a}{s - 1} + J_a(s). \quad (4.10)$$

Proposition 4.2 (A normally convergent integral for η). *Let $g_0 = \gamma_0(a)$. Then*

$$\eta(a) = J'_a(1) + \gamma B_2(a) - \frac{\gamma^2 + \zeta(2)}{2} g_0 + \frac{\gamma^3}{6} + \frac{\gamma \zeta(2)}{2} + \frac{\zeta(3)}{3}, \quad (4.11)$$

$$J_a(1) = \frac{(g_0 - \gamma)^2 + \zeta(2) - \zeta(2, a)}{2}. \quad (4.12)$$

Proof. Multiply (4.2) by the Taylor expansion of $\Gamma(1 + u)$ through cubic order and compare the constant and linear coefficients with (4.10). The expansion of $\log \Gamma(1 + u)$ gives

$$[u^2]\Gamma(1 + u) = \frac{\gamma^2 + \zeta(2)}{2}, \quad [u^3]\Gamma(1 + u) = -\frac{\gamma^3}{6} - \frac{\gamma \zeta(2)}{2} - \frac{\zeta(3)}{3},$$

which proves the two formulas. □

At $a = 1$, $\Phi(z, 1, 1) = \text{Li}_1(z)/z = -\log(1 - z)/z$ and

$$K_1(t) = -\frac{\log(1 - e^{-t})}{e^t - 1}, \quad J_1(1) = 0.$$

The integral formula reduces to

$$\begin{aligned} \eta(1) = & \frac{\gamma^3}{6} - \frac{\gamma\zeta(2)}{2} + \frac{\zeta(3)}{3} \\ & + \int_0^1 \log t \left(-\frac{\log(1-e^{-t})}{e^t-1} + \frac{\log t}{t} \right) dt + \int_1^\infty \frac{-\log t \log(1-e^{-t})}{e^t-1} dt. \end{aligned} \quad (4.13)$$

Both integrals converge ordinarily. Numerically, $\eta(1) = 0.5367870243584185675036078843488117\dots$. The numerical section compares this integral with Cauchy extraction from an independently continued ordered double sum.

All higher strict harmonic coefficients follow from the same kernel; they do not require a separate continuation. Put

$$e_j = [u^j] \frac{1}{\Gamma(1+u)}, \quad C_m(a) = [u^m] R_2(u, 0; a).$$

Multiplying (4.10) by $1/\Gamma(1+u)$ gives, for every integer $m \geq 0$,

$$C_m(a) = e_{m+2} + h_a e_{m+1} + \sum_{j=0}^m \frac{e_{m-j}}{j!} J_a^{(j)}(1), \quad (4.14)$$

$$C_m(a) = \frac{(-1)^m \gamma_H(m, a) - \zeta^{(m)}(2, a)}{m!}. \quad (4.15)$$

The second formula is (4.3) coefficient by coefficient. Each $J_a^{(j)}(1)$ is an ordinarily convergent integral obtained by inserting $\log^j t$ in (4.9). Thus the complete strict harmonic Stieltjes hierarchy has a single convergent kernel, with reciprocal-Gamma coefficients supplying its finite normalization.

4.4 Polylogarithmic primitives and harmonic coefficients

For $|z| < 1$, the strict multiple polylogarithm

$$\text{Li}_{s_1, \dots, s_d}(z) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z^{n_1}}{n_1^{s_1} \cdots n_d^{s_d}}$$

has the ordinary normalized Euler primitive

$$\int_0^z \text{Li}_{s_1, s_2, \dots, s_d}(w) \frac{dw}{w} = \text{Li}_{s_1+1, s_2, \dots, s_d}(z). \quad (4.16)$$

All fixed spectral derivatives commute with this identity on compact subdisks. Their inner coefficients are the unshifted logarithm-weighted harmonic sums that also occur in the remainder construction (2.6). At all indices one,

$$\text{Li}_{\{1\}^d}(z) = \frac{[-\log(1-z)]^d}{d!}, \quad \sum_{d \geq 1} y^{d-1} \text{Li}_{s, \{1\}^{d-1}}(z) = \sum_{n \geq 1} \frac{z^n}{n^s} \frac{\Gamma(n+y)}{\Gamma(n)\Gamma(1+y)}. \quad (4.17)$$

The second identity follows by taking elementary symmetric polynomials of $1, 1/2, \dots, 1/(n-1)$; the Gamma quotient is the polynomial $\prod_{j=1}^{n-1} (1+y/j)$, including at its apparently exceptional parameters. One may first take y near zero and then continue in y . These are classical polylogarithmic and harmonic generating identities, recorded here to give a concrete derivative and antiderivative dictionary for the Laurent coefficients. The extension at the singular endpoint is the specified spectral calculus proved above; the endpoint is not obtained by inserting $z = 1$ into a divergent series.

5 Cubic coincident Stieltjes products and a diagonal Tornheim jet

The incoming *Coincident Stieltjes* article explicitly asks for the unit-coordinate finite part of $\int_0^1 \psi(x)^3 dx$ and an all-index three-factor closure theorem. It suggests higher-depth coordinates and does not assume a reduction to ordinary zeta constants. The results below supply an exact reduction to a specified diagonal Tornheim derivative, together with a holomorphic germ for all mixed Stieltjes indices. They do not evaluate that remaining derivative in a depth-one constant algebra.

The triple-Hurwitz/Tornheim correspondence is classical, as is the Mellin–polylogarithm continuation used to compute Tornheim functions. Relevant antecedents include Espinosa–Moll, Borwein–Dilcher, Bailey–Borwein, and Onodera[8, 9, 10, 11]. The low diagonal jets below are recovered identities from this established setting.

5.1 Normalization and the classical three-factor kernel

Use the unit coordinate x and, for $e \in \mathbb{Z}$ and $k \in \mathbb{Z}_{\geq 0}$, the convention

$$\text{FP} \int_0^1 x^e \log^k x dx = \begin{cases} (-1)^k k! / (e+1)^{k+1}, & e \neq -1, \\ 0, & e = -1. \end{cases}$$

This agrees with the constant term after removing the divergent endpoint powers and logarithms from $\int_\epsilon^1$. Let

$$g(u, x) = \zeta(1+u, x) - \frac{1}{u} = \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(x) u^m, \quad b(u) = \zeta(1+u) - \frac{1}{u}, \quad a(u) = -(1+u)\zeta(2+u).$$

The removable singularities in g and b are always removed. In particular $g(0, x) = -\psi(x)$, $b(0) = \gamma$, and $a(0) = -\zeta(2)$.

Initially in its absolutely convergent region, put

$$T(A, B, C) = \sum_{m, n \geq 1} \frac{1}{m^A n^B (m+n)^C}.$$

Its meromorphic continuation is understood in every later formula. The initial absolute-convergence domain is given by $\Re(A+C) > 1$, $\Re(B+C) > 1$, and $\Re(A+B+C) > 2$. Write $K_2(s, t) = \int_0^1 \zeta(s, x) \zeta(t, x) dx$ and $K_3(s_1, s_2, s_3) = \int_0^1 \prod_{i=1}^3 \zeta(s_i, x) dx$ in an initial convergence domain, then continue meromorphically. If $\lambda_i = 1 - s_i$, the classical triple kernel is

$$K_3(s_1, s_2, s_3) = \frac{2\Gamma(\lambda_1)\Gamma(\lambda_2)\Gamma(\lambda_3)}{(2\pi)^{\lambda_1+\lambda_2+\lambda_3}} \sum_{k=1}^3 \cos \frac{\pi(\lambda_i + \lambda_j - \lambda_k)}{2} T(\lambda_i, \lambda_j, \lambda_k), \quad (\text{C1})$$

where $i < j$ are the indices other than k . For $\Re \lambda_i > 1$, the Fourier series

$$\zeta(1-\lambda, x) = \Gamma(\lambda)(2\pi)^{-\lambda} \sum_{n \neq 0} |n|^{-\lambda} e^{-\pi i \lambda \text{sgn}(n)/2} e^{2\pi i n x}$$

converges absolutely. The zero-frequency condition in the product requires two positive frequencies and one negative, or its opposite. The two sign regions paired for a fixed negative position give the cosine in (C1). This proves (C1) in a nonempty open set; meromorphic continuation gives the stated identity. The same argument gives the pair kernel

$$K_2(s, t) = \frac{2\Gamma(1-s)\Gamma(1-t)}{(2\pi)^{2-s-t}} \cos \frac{\pi(s-t)}{2} \zeta(2-s-t).$$

5.2 The complete cubic subtraction

For compactness write $K_{ij} = K_2(1+u_i, 1+u_j)$, $K_{123} = K_3(1+u_1, 1+u_2, 1+u_3)$, and $U = u_1 + u_2 + u_3$. Define the meromorphic germ

$$I_3(\mathbf{u}) = K_{123} - \sum_{k=1}^3 \frac{K_{ij}}{u_k} - \frac{1}{u_1 u_2 u_3}. \quad (\text{C2})$$

The mean of a single Hurwitz factor is zero in the initial domain, so this is precisely the meromorphic continuation of $\int_0^1 \prod_i g(u_i, x) dx$.

Theorem 5.1 (Cubic holomorphic completion). *The germ*

$$F_3(\mathbf{u}) = I_3(\mathbf{u}) + \sum_{k=1}^3 \frac{a(u_k)}{u_i + u_j} + \sum_{k=1}^3 \frac{b(u_i)b(u_j)}{u_k} \quad (\text{C3})$$

is holomorphic in a neighborhood of $\mathbf{u} = 0$. For all $m_1, m_2, m_3 \geq 0$,

$$\text{FP} \int_0^1 \prod_{i=1}^3 \gamma_{m_i}(x) dx = (-1)^{m_1+m_2+m_3} \prod_{i=1}^3 m_i! [u_1^{m_1} u_2^{m_2} u_3^{m_3}] F_3(\mathbf{u}). \quad (\text{C4})$$

Proof. The shift relation gives

$$g(u, x) = x^{-1-u} + h(u, x), \quad h(u, x) = g(u, 1+x), \quad h(u, x) = b(u) + a(u)x + O(x^2).$$

All these Taylor statements are uniform on small compact spectral neighborhoods. Write $h_i = h(u_i, x)$. The product of the three singular terms contributes $-1/(2+U)$ after meromorphic integration. A term with the two singular factors in positions i, j and smooth factor h_k contributes

$$-\frac{b(u_k)}{1+u_i+u_j} - \frac{a(u_k)}{u_i+u_j} + \int_0^1 x^{-2-u_i-u_j} (h_k - b(u_k) - a(u_k)x) dx.$$

A term with just the singular factor in position k contributes

$$-\frac{b(u_i)b(u_j)}{u_k} + \int_0^1 x^{-1-u_k} (h_i h_j - b(u_i)b(u_j)) dx.$$

The remaining term is $\int_0^1 h_1 h_2 h_3 dx$. After (C3), all remaining integrals are normally convergent near zero: their endpoint majorants have the form $Cx^{-\eta}(1+|\log x|^M)$, with $\eta < 1$, after any prescribed finite number of spectral derivatives. This proves holomorphy.

The deleted expressions are exactly the integrations of resonant monomials with exponent -1 at $\mathbf{u} = 0$. Each coefficient of such a monomial is a multiple of $x^{-1} \log^j x$, whose unit-coordinate finite part is zero. The other integrated monomials have precisely the finite parts prescribed above. Coefficient extraction under the normally convergent remainder proves (C4). $\square$

The mixed spectral derivatives in (C4) apply to the completed germ. Uncompleted individual Tornheim summands can have intersecting polar hyperplanes at the origin. In particular one must not read (C4) as a formula involving unspecified ordinary partial derivatives of T at $(0, 0, 0)$.

5.3 Every argument derivative

The same proof gives a finite all-order extension. For $p_i \in \mathbb{Z}_{\geq 0}$, let

$$\begin{aligned} c_p(u) &= (-1)^p(1+u)_p, & h_i(u_i, x) &= \partial_x^{p_i} g(u_i, 1+x), \\ G_i &= \partial_x^{p_i} g(u_i, x) = c_{p_i}(u_i) x^{-p_i-1-u_i} + h_i(u_i, x). \end{aligned}$$

Let I_p be the meromorphic product integral of the G_i . With $s_i = 1 + p_i + u_i$, it is explicitly

$$\begin{aligned} I_p &= \left(\prod_{i=1}^3 c_{p_i}(u_i) \right) K_3(s_1, s_2, s_3) \\ &\quad - \sum_{k=1}^3 \frac{\delta_{p_k 0} c_{p_i}(u_i) c_{p_j}(u_j)}{u_k} K_2(s_i, s_j) - \frac{\prod_{i=1}^3 \delta_{p_i 0}}{u_1 u_2 u_3}. \end{aligned}$$

The single-Hurwitz integrals vanish in the initial convergence domain and hence identically after continuation. Then the holomorphic completion is

$$\begin{aligned} F_p &= I_p + \sum_{k=1}^3 \frac{c_{p_k}(u_k)}{u_k} [x^{p_k}] (h_i(u_i, x) h_j(u_j, x)) \\ &\quad + \sum_{k=1}^3 \frac{c_{p_i}(u_i) c_{p_j}(u_j)}{u_i + u_j} [x^{p_i+p_j+1}] h_k(u_k, x). \end{aligned} \tag{C5}$$

Its coefficients generate FP $\int_0^1 \prod_i \gamma_{m_i}^{(p_i)}(x) dx$ with the same factorial and sign in (C4). Indeed a subset S of singular factors has exponent $-|S| - \sum_{i \in S} p_i - \sum_{i \in S} u_i$; the unique resonant smooth Taylor degree is $|S| + \sum_{i \in S} p_i - 1$. For a singleton or pair this gives precisely (C5). For all three factors the needed smooth degree is positive but there is no smooth factor, so no further resonant term occurs. To justify the normal convergence rather than merely identify the formal poles, subtract every smooth Taylor term up to that threshold. The integrated lower Taylor degrees have nonzero integer denominators at the origin, so they are holomorphic and retain their prescribed monomial finite parts. The only denominators vanishing at the origin are exactly those displayed in (C5). The remaining Taylor remainders are integrable uniformly on sufficiently small compact spectral neighborhoods, also after any fixed finite number of derivatives. Every smooth coefficient is a finite combination of Hurwitz jets, because

$$[x^\ell] h_i(u_i, x) = \frac{c_{p_i+\ell}(u_i)}{\ell!} \zeta(1 + p_i + \ell + u_i) - \frac{\delta_{p_i 0} \delta_{\ell 0}}{u_i}.$$

The Pochhammer coefficient expansions are the usual finite harmonic polynomials. Thus (C5) is an explicit finite recipe for all cubic argument-derivative moments.

5.4 An independently defined diagonal Tornheim function

Let $\tau(s) = T(s, s, s)$ initially for $\Re s > 2/3$, and then continue this one-variable function. The diagonal restriction is essential: T itself has no direction-independent value at the multivariate origin. For example, $T(s, s, 0) = \zeta(s)^2$ tends to $1/4$, whereas the diagonal limit is $1/3$, as recovered in (C8) below.

Here is a convergent formula defining the germ of τ near zero without using any Hurwitz product integral. Set $L_s(t) = \text{Li}_s(e^{-t})$, $G(s) = \Gamma(1-s)$, and

$$\begin{aligned} R_s(t) &= L_s(t)^2 - G(s)^2 t^{2s-2} - 2G(s)\zeta(s)t^{s-1} + 2G(s)\zeta(s-1)t^s - \zeta(s)^2, \\ E(s) &= \int_0^1 t^{s-1} R_s(t) dt + \int_1^\infty t^{s-1} L_s(t)^2 dt. \end{aligned}$$

For sufficiently small $\delta > 0$, E is holomorphic when $|s| < \delta$. Indeed the Jonquière expansion [14]

$$L_s(t) = \Gamma(1-s)t^{s-1} + \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \zeta(s-k)t^k, \quad |t| < 2\pi,$$

shows that the subtracted integrand is bounded by $C(t^{-2\delta} + t^{-\delta})(1 + |\log t|^M)$ after any fixed number of spectral derivatives. The second integral converges exponentially. Direct Mellin integration first for $2/3 < \Re s < 1$ and then continuation gives the holomorphic formula

$$\tau(s) = \frac{1}{\Gamma(1+s)} \left\{ sE(s) + \frac{s\Gamma(1-s)^2}{3s-2} + \frac{2s\Gamma(1-s)\zeta(s)}{2s-1} - \Gamma(1-s)\zeta(s-1) + \zeta(s)^2 \right\}. \quad (\text{C6})$$

This is the $\theta = 1$ local-subtraction form of the classical Crandall Mellin continuation. In particular $\tau'''(0)$ is an unambiguous, independently computable spectral constant.

5.5 The cubic digamma reduction

Put $\ell = \log(2\pi)$, and define

$$Z(t) = t\zeta(1+t), \quad A_d(t) = \frac{\Gamma(1-t)^2}{\Gamma(1-2t)\cos(\pi t)}, \quad B(t) = \Gamma(1-t)^3(2\pi)^{3t}\cos(\pi t/2).$$

The classical pair formula yields $K_2(1+t, 1+t) = -A_d(t)Z(2t)/t^2$, while (C1) gives $K_3(1+t, 1+t, 1+t) = -6B(t)\tau(-t)/t^3$. Hence

$$F_3(t, t, t) = \frac{-6B(t)\tau(-t) + 3A_d(t)Z(2t) - 1 + 3(Z(t) - 1)^2}{t^3} + \frac{3a(t)}{2t}. \quad (\text{C7})$$

Since the left side is holomorphic, comparison of the three negative powers gives

$$\tau(0) = \frac{1}{3}, \quad \tau'(0) = \ell, \quad \tau''(0) = 3\ell^2 - 2\gamma^2 - 4\gamma_1 + \zeta(2). \quad (\text{C8})$$

These are recovered low-jet identities; the first two are well known, and Onodera's 2021 paper evaluates second derivatives at the origin.

For completeness the needed expansions are

$$\begin{aligned} Z(t) &= 1 + \gamma t - \gamma_1 t^2 + \frac{1}{2}\gamma_2 t^3 + O(t^4), \\ A_d(t) &= 1 + 2\zeta(2)t^2 - 2\zeta(3)t^3 + O(t^4), \\ \log B(t) &= 3(\gamma + \ell)t + \frac{3}{4}\zeta(2)t^2 + \zeta(3)t^3 + O(t^4), \\ a'(0) &= -\zeta(2) - \zeta'(2). \end{aligned}$$

Taking the constant coefficient of (C7) and using $F_3(0, 0, 0) = -\text{FP} \int_0^1 \psi(x)^3 dx$ proves

$$\begin{aligned} \text{FP} \int_0^1 \psi(x)^3 dx = & -\tau'''(0) - 9\gamma^3 - 18\gamma^2\ell - 30\gamma\gamma_1 - 36\ell\gamma_1 - 12\gamma_2 + 9\ell^3 \\ & + \frac{3}{2}\gamma\zeta(2) + 9\ell\zeta(2) + \frac{3}{2}\zeta(2) + 8\zeta(3) + \frac{3}{2}\zeta'(2). \end{aligned} \quad (\text{C9})$$

An equivalent unsimplified coefficient certificate is

$$\begin{aligned} F_3(0, 0, 0) = & -6[t^3]B(t)\tau(-t) + 12\gamma_2 + 12\gamma\zeta(2) - 6\zeta(3) - 6\gamma\gamma_1 \\ & - \frac{3}{2}(\zeta(2) + \zeta'(2)). \end{aligned}$$

5.6 Directional Laurent constants do not fix the coordinate finite part

Let $\alpha_1, \alpha_2, \alpha_3$ be fixed complex numbers with $\alpha_i \neq 0$ and $\alpha_i + \alpha_j \neq 0$. If CT denotes the one-variable Laurent constant and $Q = \text{FP} \int_0^1 \gamma_0(x)^3 dx$, then (C3) gives

$$\begin{aligned} Q = & \text{CT}_{t=0} I_3(\alpha_1 t, \alpha_2 t, \alpha_3 t) - (\zeta(2) + \zeta'(2)) \sum_{k=1}^3 \frac{\alpha_k}{\alpha_i + \alpha_j} \\ & - \gamma\gamma_1 \sum_{k=1}^3 \frac{\alpha_i + \alpha_j}{\alpha_k}. \end{aligned} \quad (\text{C10})$$

For the diagonal this correction is $-\frac{3}{2}(\zeta(2) + \zeta'(2)) - 6\gamma\gamma_1$. This is a normalization warning, not an identified false claim in an inspected source. Removing only the negative Laurent powers after diagonal specialization would omit the finite coefficients of the resonant expressions. The full germ (C3) removes exactly those entire expressions and yields the fixed-coordinate finite part.

5.7 Independent numerical diagnostic

The script `verify_cubic.py` computes the first four diagonal Tornheim derivatives from (C6), using the convergent Jonquière series on $(0, 1)$ and an exponentially convergent integral on $(1, \infty)$. The independent digamma calculation integrates

$$\psi(x)^3 + x^{-3} + 3\gamma x^{-2} - 3(\zeta(2) - \gamma^2)x^{-1}$$

ordinarily and then adds $1/2 + 3\gamma$. Near zero it evaluates the regular remainder by the convergent Taylor series for $\psi(1+x)$, avoiding cancellation of huge singular terms. At 35 working decimal digits and 50 Jonquière terms the values are

$$\begin{aligned} \tau'''(0) & \approx 86.657866601100073652903276858762183, \\ \text{FP} \int_0^1 \psi(x)^3 dx & \approx 1.9662627343915343406643753384116314. \end{aligned}$$

The two independent expressions in (C9) differ by less than 4.1×10^{-34} . This is a floating-point diagnostic, not an interval enclosure or substitute for the proof above.

6 Verification, proposed status updates, and integration

6.1 What the proofs establish and what the computations check

The analytic arguments establish the normal convergence of the tail remainders, the meromorphic identities, the exact curve-jet bound, and the endpoint completion at every stated index. These conclusions do not depend on numerical precision. The attached verification has two distinct roles: exact algebra checks the printed coefficients, and independent numerical constructions test signs, normalizations, and representative specializations.

The primary exact calculation uses Newton’s elementary-symmetric recurrence with a separate formal symbol for every Hurwitz jet. It does not obtain the displayed coefficients by differentiating a copy of the displayed exponential. Sixty checks cover the Gamma polynomials $B_1, \dots, B_5$, the explicit formulas through depth five, the equal-slope Laurent coefficients through depth seven, the zero-trailing slope family, and the complete six-permutation depth-three stuffle identity. A separately written calculation performs seven additional ordered-coefficient checks. Five further exact checks verify the cubic pole cancellations, the complete constant in (C9), and the Mellin counterterm exponents. The seventy-two checks all pass.

For the ordered numerical diagnostic, the double and triple sums are continued directly by summing a finite initial range and attaching Euler–Maclaurin tails of the ordinary Hurwitz factors. Cauchy coefficient extraction then obtains the finite part on a small complex circle. This calculation does not use the regular-germ recursion to evaluate the left side. At $a = 1$, the additional coefficient η is also computed from the independently convergent integral (4.13). The cubic-product diagnostic separately uses the polylogarithmic Mellin formula and the spatially subtracted digamma integral described in Section 5.

Diagnostic	Representative value
$\eta(1)$	0.536787024358418567503607884349...
$\mathcal{F}_3(1, \frac{1}{2}, 1)$	0.446979185190867817804303675619...
FP $Z_3(1 + \frac{5}{4}\varepsilon, 1 + \frac{1}{2}\varepsilon, 1 + \frac{3}{4}\varepsilon; \frac{3}{4})$	0.169462871179343781164830292059...
$\tau'''(0)$	86.6578666011000736529032768588...
FP $\int_0^1 \psi(x)^3 dx$	1.96626273439153434066437533841...

The recorded ordered run uses 50 working decimal digits, twenty Cauchy nodes, a finite cutoff of 36, and twenty Euler–Maclaurin terms. The direct triple evaluation uses forty additional guard digits to protect the small high-order Hurwitz tails before their multiplication by large Euler–Maclaurin coefficients. Across the four depth-three comparisons at $a = 1$ and $a = 3/4$, the largest absolute residual is below 8×10^{-35} ; the independent $\eta(1)$ comparison is below 10^{-38} . The digamma-cube comparison uses 35 working digits and fifty Jonquière terms, with residual below 4.1×10^{-34} . The saved JSON records the actual complex values and residuals. These are floating-point diagnostics, with truncation and roundoff; the number of working digits is not a claim that every printed digit is certified. No integer-relation search is used as proof, and no interval enclosure is asserted.

6.2 Independent review and limits of the source audit

The ordered and cubic derivations were reviewed independently of their initial derivation, including the tail majorants, divided differences, factorials, Fourier phases, endpoint resonances, and the

third-order Tornheim expansion. A separate review checked the sharp curve perturbation, the slope residue, the harmonic conversion, and the Mellin kernel. Review led to a more explicit argument-derivative kernel: both the triple and pair kernels carry their shifted spectral arguments. It also led to an explicit remainder subtraction through the full integrability threshold. Those clarifications are incorporated in the proofs, rather than left as qualifications to the formulas.

No new concrete false mathematical claim was identified in the inspected canonical or incoming material. In particular, the inspected quadratic coincident completion and the recent Dougall resonance formulas were consistent with their stated scope. This is a targeted audit, not a claim that the hundreds of pages in the canonical manuscript have all been re-proved. Several tempting proposed extensions were already present in the incoming reports and have not been relabelled as new results here.

The changes recommended for the repository are therefore precise status and normalization updates:

1. Mark the independent-exponent ordered depth-three question in the resonant-jets report as answered by Theorems 2.1 and 2.4. Record the stronger all-depth specialization in Theorem 2.3.
2. Mark its tangential-curve question as answered by Theorem 3.1, retaining the hypothesis that no prefix is identically zero. The required jet order is a theorem with a sharpness construction, not an empirical order suggested by examples.
3. Mark the cubic coincident-moment target as solved in specified Tornheim coordinates by (C3)–(C6), and the digamma cube as reduced by (C9). Keep the reduction of $\tau'''(0)$ to a chosen depth-one algebra as an open question.
4. Cite the established Laurent work of Saha and of Matsumoto, Onozuka and Wakabayashi; preserve the harmonic convention conversion (4.4). Cite the classical triple-Hurwitz and Mellin–Tornheim inputs when integrating the cubic section.
5. Distinguish the spatial finite part, the regularized multivariate germ, and the constant on a specified spectral curve. Formula (C10) gives their concrete cubic comparison. It should be added as a normalization identity, not announced as an erratum to a source that has not been shown to confuse them.
6. Preserve the conjectural status of the short S_6 and revised S_8 candidates. Their existing proximity certificates and restricted separation results are not proofs or general disproofs of the proposed period identities.

6.3 Reproducible package

The package contains the modular article source, a flattened standalone TeX source, the compiled PDF, exact-check programs, independent numerical programs, recorded JSON results, pinned source hashes, and a claim-by-claim integration ledger. The proofs are ordinary mathematical proofs; no Lean or other proof-assistant formalization is claimed. The build script compiles the article without fetching any mathematical data. Verification uses Python, SymPy and mpmath; its commands and recorded versions are in the package README.

The proposed placement follows the repository’s existing thematic research-report structure. The delivered section sources can also be integrated into the canonical manuscript after notation

reconciliation. The present delivery does not modify the repository or promote its unrelated conjectures.

7 Further research questions and exact-identity programme

The following questions begin where the proved formulas stop. They are proposed research directions, not assertions disguised as conjectures. Their intended outputs are exact identities, finite coordinate descriptions, or proofs of specific candidate equalities.

7.1 Independent slopes beyond depth three

Q1. The first genuinely new ordered coefficients. At depth four the finite part of a general ray uses the linear jet of R_3 and the quadratic jet of R_2 , in addition to ordinary Stieltjes data. Determine a complete finite basis of these coefficients modulo the exact stuffle relations, shift identities, and the known diagonal Gamma constraints. Then write a formula analogous to (2.18) for all transverse quadruples of slopes. The first objective is a proved spanning description; any claim of arithmetic independence would require a separate argument.

Q2. Classify the directions with ordinary-zeta closure. The family with equal trailing slopes has ordinary Hurwitz closure at every depth. At depth three the coefficient of the ordering coordinate is $(c_2 - c_3)/c_1$. At depth four and beyond, determine the algebraic loci on which all genuinely nonsymmetric coefficient contributions cancel. One can first work in the formal quotient by proved identities, where the question becomes an exact finite linear-algebra problem. Accidental numerical relations at a specific shift must be distinguished from identities valid as functions of a .

Q3. Several blocks of equal slopes. Replace $(p, q, \dots, q)$ by a sequence of several constant blocks. Does each fixed number of blocks admit a generating function involving a finite collection of lower-depth regular germs, uniformly in the block lengths? The all-equal and one-distinguished-slope cases suggest an induction by ordered blocks, but the crossing terms need explicit control. A useful deliverable would be an exact two-block formula at all depths and a proof of its coordinate requirements.

7.2 Harmonic, polylogarithmic, and Gamma relations

Q4. Rational-shift distribution of the ordering coordinate. The shift relation (4.5) is exact. The existing root-of-unity filter in the incoming endpoint report already gives finite coloured descriptions of rational-shift ordered sums and their fixed spectral derivatives [4]. The harder question is which character-weighted combinations of $\eta(r/q)$ eliminate the residual coloured ordering coefficients and reduce entirely to ordinary Hurwitz or Dirichlet L -derivatives. Determine the exact elimination spaces first at conductors 2, 3, 4, 6, then seek a conductor-uniform rule. The inclusive harmonic literature [7] is a starting point; the strict/inclusive conversion must remain explicit.

Q5. Normalized parameter primitives. Equation (4.6) reduces $\eta'(a)$ to absolutely convergent ordered sums. Seek a useful normalized antiderivative in the shift parameter, expressed through multiple Gamma or Barnes-type functions and spectral zeta derivatives. A candidate must specify its

additive constant and prove differentiation back to η . The whole higher harmonic hierarchy is already represented by (4.14)–(4.15); seek a parameter-primitive relation compatible with that hierarchy and with the exact shift recurrence. Any regularized integral must carry its finite normalization explicitly.

Q6. Logarithmic decorations of convergent primitive families. One Euler integration raises the outer order from one to two, so $Z_{r+1}(2 + u_0, 1 + u_1, \dots, 1 + u_r; a)$ is holomorphic near the origin. For this ordinary primitive the endpoint limit and every fixed spectral derivative commute by domination; there is no unexplained regularization defect. The incoming endpoint work already evaluates the unweighted integer-shift primitive hierarchy [4]. The next target is to classify which independent logarithmic decorations, obtained by differentiating the separate indices, still reduce to ordinary Gamma, zeta and Stieltjes jets. Begin with the convergent depth-two derivatives appearing in (4.6), derive their exact integral relations, and compare them with the height-one Gamma generating relations.

7.3 Tornheim derivatives and higher coincident products

Q7. Evaluate the remaining cubic Tornheim derivative. Can $\tau'''(0)$ be reduced to a specified algebra generated by γ_m , ordinary $\zeta^{(j)}(n)$, logarithms, and perhaps Barnes Gamma constants? Formula (C9) turns this into an equivalent exact evaluation question for the coordinate-normalized digamma cube. The independent Mellin representation permits systematic analytic manipulations and high-precision exploration, but an integer relation would remain a candidate until a functional identity proves it. Existing Tornheim derivative work and the recent Kronecker-limit literature should be checked before any priority claim [9, 10, 11, 12].

Q8. Minimal coordinates for all cubic moments. Formula (C4) is a complete generating prescription. It does not yet classify how many independent completed Tornheim jets are needed at each total Stieltjes index $m_1 + m_2 + m_3$. Determine exact reductions under permutation symmetry, integration by parts with the stated finite-part convention, and Tornheim functional equations. The first useful target is a table for total indices through three, with proofs and a transparent distinction between a spanning family and an arithmetically minimal one.

Q9. Four and more coincident factors. Extend the cubic Fourier kernel to the zero-frequency sums arising from four Hurwitz factors. The endpoint resonances can be organized by subsets of singular factors and their smooth Taylor degrees; the genuinely new work is the global multiple-sum coordinate and its functional reductions. Begin with a fixed-coordinate formula for FP $\int_0^1 \psi(x)^4 dx$, keeping all spectral-counterterm finite contributions. The natural higher-depth function should be defined independently, as (C6) defines τ here.

7.4 Singular geometry and the Gaussian programme

Q10. Restrictions lying inside a polar hyperplane. Theorem 3.1 handles every analytic curve whose prefixes are not identically zero. A curve contained in $P_j = 0$ needs an additional prescription. Compare residue-first, regular-germ-first, and finite-part-in-a-normal-variable constructions, and derive their exact differences. Decide which prescriptions respect the stuffle product and how those

choices propagate under changes of coordinates. This should be formulated as a comparison theorem; no canonical value follows just from substituting into a rational slope formula.

Q11. A functional proof of the short Gaussian candidates. The current S_6 and revised S_8 candidates require identities between specified polylogarithmic periods. A productive next step would be an exact functional relation, differential equation with enough boundary data, or a finite symbolic certificate in a rigorously defined iterated-integral algebra. Existing restricted separator results constrain particular ansatz classes; they do not establish general period independence. The ordered Hurwitz identities in this article may supply useful coefficient relations, but no demonstrated bridge presently converts them into a proof of either Gaussian candidate. That bridge itself should be stated as a concrete research target rather than assumed.

References

- [1] V. Reshetnikov and the ProveIt research programme, *Polylogarithms and their Arithmetic Bridges*, canonical manuscript and status records, snapshot 4c1286161c42, 10 October 2026; full commit identifier in Section 1. [Pinned manuscript directory](#).
- [2] ProveIt research continuation, *Resonant Gamma Jets and Directional Zeta Identities: Discrete Stieltjes products, shifted polylogarithms, and local formulas for higher spectral derivatives*, 10 October 2026. Incoming archive ProveIt_Resonant_Jets_2026-10-10.zip, sections/08-audit-research.tex, further-research questions Q1 and Q3. [Pinned incoming directory](#).
- [3] ProveIt research continuation, *Coincident-Point Stieltjes Calculus: Exact Quadratic Closure, Polygamma Parity, and Polylogarithmic Collision Subtraction*, 10 October 2026. Incoming archive ProveIt_Coincident_Stieltjes_2026-10-10.zip, article/article.tex, subsection “Cubic coincident moments and higher-depth coordinates.”
- [4] ProveIt research continuation, *Endpoint Regularization of Logarithmic Harmonic Polylogarithms: Stieltjes coefficients, Gamma transfer, spectral finite parts, and exact antiderivatives*, 10 October 2026. Incoming archive ProveIt_Endpoint_Regularization_2026-10-10.zip, article/endpoint_regularization.tex.
- [5] B. Saha, Multiple Stieltjes constants and Laurent type expansion of the multiple zeta functions at integer points, *Selecta Math. (N.S.)* **28** (2022), article 6. <https://doi.org/10.1007/s00029-021-00719-1>. Author preprint: <https://arxiv.org/abs/1902.04389>.
- [6] K. Matsumoto, T. Onozuka and I. Wakabayashi, Laurent series expansions of multiple zeta-functions of Euler–Zagier type at integer points, *Math. Z.* **295** (2020), 623–642. <https://doi.org/10.1007/s00209-019-02337-2>. Author preprint: <https://arxiv.org/abs/1601.05918>.
- [7] L. Kargin, A. Dil, M. Cenkci and M. Can, On the Stieltjes constants with respect to harmonic zeta functions, *J. Math. Anal. Appl.* **525** (2023), article 127302. <https://doi.org/10.1016/j.jmaa.2023.127302>. Author preprint: <https://arxiv.org/abs/2304.03517>.
- [8] O. Espinosa and V. H. Moll, The evaluation of Tornheim double sums. Part 1, *J. Number Theory* **116** (2006), 200–229. <https://doi.org/10.1016/j.jnt.2005.04.008>. Author preprint: <https://arxiv.org/abs/math/0505647>.

- [9] J. M. Borwein and K. Dilcher, Derivatives and fast evaluation of the Tornheim zeta function, *Ramanujan J.* **45** (2018), 413–432. <https://doi.org/10.1007/s11139-017-9890-9>.
- [10] D. H. Bailey and J. M. Borwein, Computation and experimental evaluation of Mordell–Tornheim–Witten sum derivatives, *Exp. Math.* **27** (2018), 370–376. <https://doi.org/10.1080/10586458.2017.1295687>. Author preprint: <https://www.davidhbailey.com/dhbpapers/omega-numerics.pdf>.
- [11] K. Onodera, On multiple Hurwitz zeta function of Mordell–Tornheim type, *Int. J. Number Theory* **17** (2021), 2327–2360. <https://doi.org/10.1142/S1793042121500913>.
- [12] S. Sathyanarayana and N. G. Sharan, Mordell–Tornheim zeta function: Kronecker limit type formulas and special values, *Adv. Appl. Math.* **177** (2026), article 103075. <https://doi.org/10.1016/j.aam.2026.103075>. Author preprint: <https://arxiv.org/abs/2510.10093>.
- [13] NIST Digital Library of Mathematical Functions, Section 25.11, Hurwitz zeta function. <https://dlmf.nist.gov/25.11>. Accessed 10 October 2026.
- [14] NIST Digital Library of Mathematical Functions, Section 25.12, Polylogarithms. <https://dlmf.nist.gov/25.12>. Accessed 10 October 2026.